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Internship : Data Analytics using Python
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Task 131 : Create a user-friendly interface for decision-making.

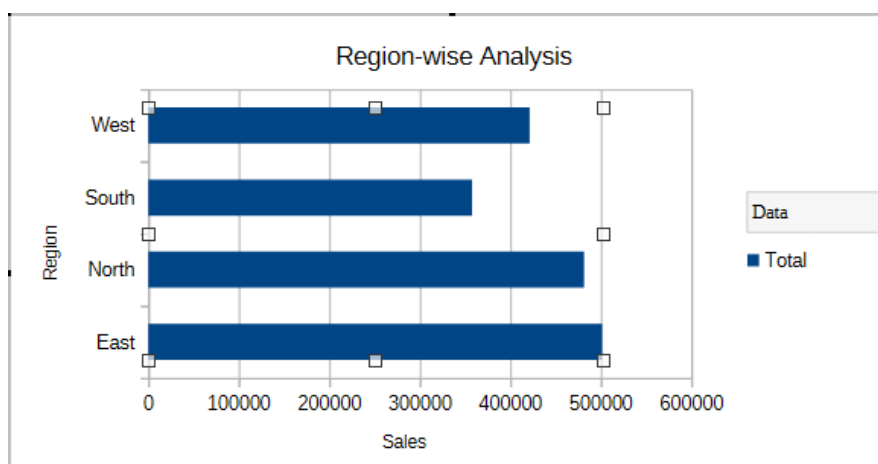
Objective

To design a user-friendly and interactive dashboard that presents data through pivot tables, visualizations, and formatted metrics, enabling users to easily analyze information and make informed decisions.

1.Region-wise Sales(Using Bar Chart) :

a) Creation of Pivot Table :

Region	Sum - Sales
East	501428
North	481407
South	357687
West	421345
Total Result	1761867



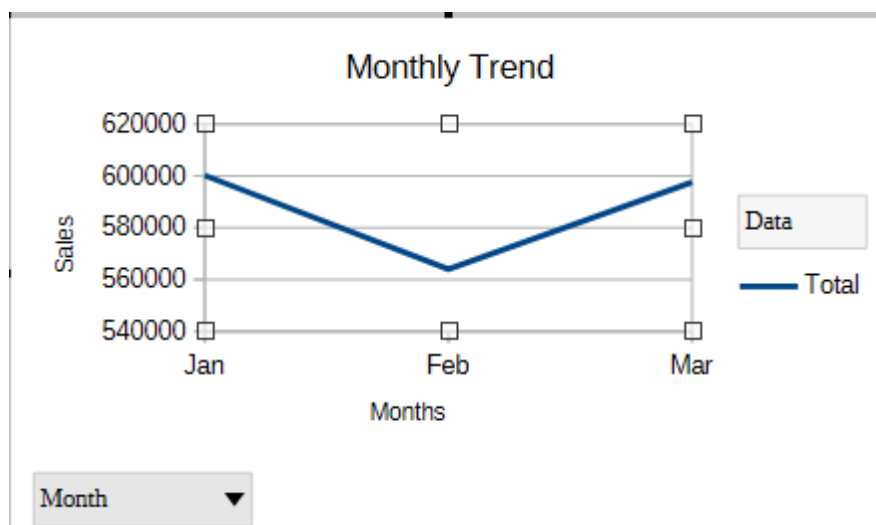
Meaningful Insights:

- The **East region** has the **highest sales (~500,000)**, making it the top-performing region.
- The **North region** follows closely with around **480,000**, showing strong performance.
- The **West region** has moderate sales (~420,000).
- The **South region** records the **lowest sales (~350,000)**.

2. Monthly Trend(Using Line Chart) :

b) Creation of Pivot Table :

Month	Sum - Sales
Jan	600179
Feb	564031
Mar	597657
Total Result	1761867



Insights :

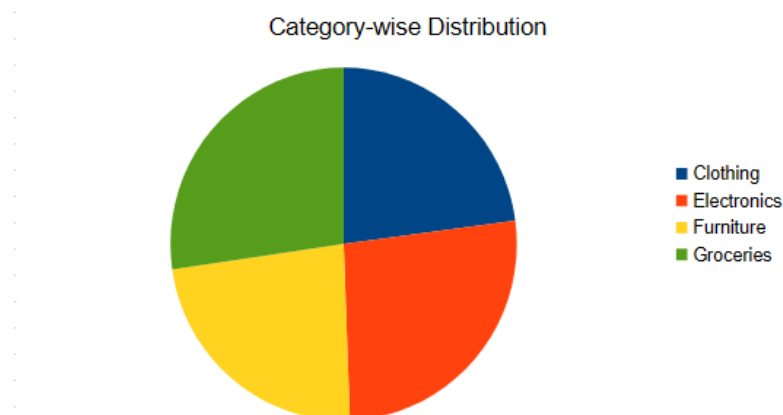
1. Peak Performance in January

- January has the highest sales
- Suggests peak demand or successful strategies during this period

3. Product Distribution(Pie-chart) :

c) Creation of Pivot Table :

Product Category	Sum - Sales
Clothing	₹403,067.00
Electronics	₹467,691.00
Furniture	₹410,046.00
Groceries	₹481,063.00
Total Result	₹1,761,867.00



Insights :

- Groceries has the highest sales (₹4,81,063)
- Electronics is the second highest contributor
- Furniture and Clothing show moderate sales
- Sales are evenly distributed across categories
- Total sales are strong (₹17,61,867)

I created a user-friendly decision-making interface in Excel by designing an interactive dashboard.

I used pivot tables to analyze data and developed visualizations for region-wise sales, monthly sales trends, and product-wise distribution.

These visuals help users easily understand performance across different areas. The dashboard enables quick comparison and supports effective decision-making.

Task 132 : Prepare explanations and documentation for collaborators.

Objective:

To prepare clear explanations and structured documentation of the data analysis process, including data cleaning, pivot table creation, and visualizations, so that collaborators can easily understand, use, and reproduce the work.

Sales Data Analysis Dashboard Documentation

1 . Introduction :

This document explains the data analysis and dashboard created using Excel. The goal is to provide a user-friendly interface for analyzing sales data and supporting decision-making.

2. Data Source :

The dataset used is a sales dataset containing:

- Date
- Region
- Product Category
- Sales
- The data represents sales transactions across different regions and categories.

3. Data Cleaning :

The following data cleaning steps were performed:

- Removed duplicate records
- Handled missing values
- Formatted the Date column correctly
- Converted Sales values into currency format (₹)
- Ensured consistent naming for regions and categories

4. Data Analysis (Pivot Tables) :

Pivot tables were created to summarize the data:

Region-wise Sales → To compare performance across regions

Monthly Sales Trend → To analyze sales over time

Product Category Distribution → To understand contribution of each category

5. Visualizations :

Charts were created for better understanding:

- Bar Chart → Region-wise Sales
- Line Chart → Monthly Sales Trend
- Pie Chart → Product Category Distribution

These visualizations help users quickly identify patterns and trends.

Task 133 : Provide resources for users to use the data analysis system .

Objective

To provide necessary resources such as dataset, dashboard, and user instructions so that users can easily access, understand, and use the data analysis system.

Resources for Data Analysis System

1. Dataset Files

- The original sales dataset is provided in Excel format
- Contains columns: Date, Region, Product Category, Sales
- Cleaned and formatted for analysis

2. Dashboard / Report

An interactive Excel dashboard is created

Includes:

- Region-wise sales analysis
- Monthly sales trend
- Product category distribution
- Pivot tables and charts are used for summarization

3. User Guide

How to Use the Dashboard

1. Open the Excel dashboard file

2. Use slicers (filters) to select:

- Date
- Region
- Product Category

3. Charts will update automatically based on selection

✳ How to Read Charts

- Bar Chart → Compare region-wise sales
- Line Chart → Understand monthly trends
- Pie Chart → View product category distribution

4. System Usage Instructions

- Ensure Excel is enabled with pivot table support
- Do not modify pivot structure to avoid errors
- Refresh data if changes are made

5. Resource Sharing

Files can be shared via:

- Google Drive (for easy access)
- GitHub (for version control)

Task 134: Improve efficiency in Excel .

Objective

To improve efficiency in Excel by using advanced formulas, pivot tables, conditional formatting, and keyboard shortcuts to perform faster data analysis and automate repetitive tasks.

Excel Learning & Efficiency Improvement

1. Skills Learned

Basic Excel :

- Data entry and formatting
- Sorting and filtering

2. Data Analysis

► Pivot Tables :

- Created pivot tables for data summarization
- Applied filters like Date, Region, and Category

3. Data Visualization

► Created charts:

- Region-wise sales (Bar Chart)
- Monthly trend (Line Chart)
- Product category distribution (Pie Chart)

4. Efficiency Techniques

- Used shortcuts like Ctrl + Shift + L
- Reduced manual work using formulas
- Organized data for faster analysis

5. Outcome

- Improved speed and accuracy
- Created interactive dashboard
- Enabled better decision-making

Task 135 : Learn SQL queries and access databases .

Objective:

To learn and apply SQL queries for accessing and managing databases, including retrieving, filtering, grouping, and analyzing data using commands like SELECT, WHERE, GROUP BY, ORDER BY, and JOIN.

```
# 135 Learn SQL queries and access databases.
import sqlite3
conn=sqlite3.connect("s.db")
cursor=conn.cursor()
cursor.execute(""" Create table Sales_data(region text,product text,sales integer)""")
cursor.execute("Insert into Sales_data values('West','laptop',50000)")
cursor.execute("Insert into Sales_data values('East','Mobile',30000)")
cursor.execute("Insert into Sales_data values('West','Tablet',40000)")
conn.commit()

# SELECT
cursor.execute("Select * from Sales_data")
print(cursor.fetchall())

[('West', 'laptop', 50000), ('East', 'Mobile', 30000), ('West', 'Tablet', 40000)]

# WHERE
cursor.execute("Select * from Sales_data where region='West'")
print(cursor.fetchall())

[('West', 'laptop', 50000), ('West', 'Tablet', 40000)]

# GROUPBY
cursor.execute("Select region,SUM(sales) from Sales_data GROUP BY region")
print(cursor.fetchall())

[('East', 30000), ('West', 90000)]
```

I created a SQLite database using Python and built a table named Sales_data to store region, product, and sales information.

I inserted sample records into the table and performed basic SQL queries such as SELECT to retrieve all data, WHERE to filter records for a specific region (West), and GROUP BY to summarize total sales by region.

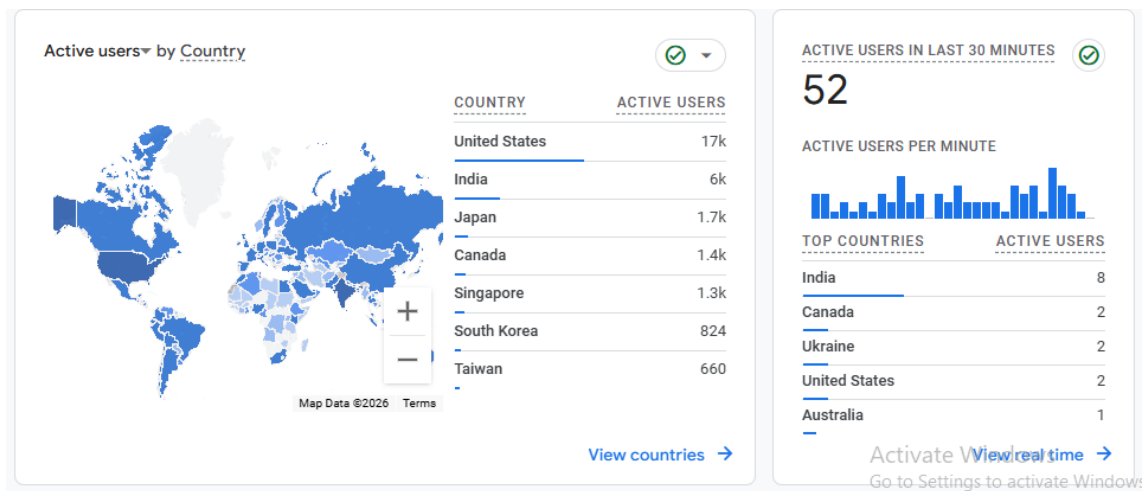
This helped me understand how to access, filter, and analyze data using SQL queries within a database system.

Task 136 : Use Analytics Tools like Google Analytics .

Objective

To collect, measure, and analyze user data in order to understand website/app performance, track user behavior, and make data-driven decisions for improving traffic, engagement, and overall business growth.

1. User attributes overview



2. Traffic acquisition

	Session primary...channel group)	Sessions	Engaged sessions	Engagement rate	Average engagement time per session
✓	Total	97,420 100% of total	33,843 100% of total	34.74% Avg 0%	41s Avg 0%
✓	1 Direct	62,945 (64.61%)	13,354 (39.46%)	21.22%	26s
✓	2 Organic Search	19,175 (19.68%)	12,680 (37.47%)	66.13%	1m 00s
✓	3 Unassigned	5,961 (6.12%)	534 (1.58%)	8.96%	48s
✓	4 Cross-network	4,608 (4.73%)	3,824 (11.3%)	82.99%	1m 42s
✓	5 Paid Search	2,685 (2.76%)	1,807 (5.34%)	67.3%	1m 22s

Activate Window →
Go to Settings to activate Window

I used the Google Analytics Demo Account to understand how analytics tools work. I selected the Google Merchandise Store (Demo) property and explored various reports.

I analyzed the User Attributes Overview to understand user demographics and behavior.

I also studied the Traffic Acquisition report to identify different traffic sources such as organic search, direct visits, and referrals. This helped me understand how users interact with a website and how data can be used to improve performance.

Task 137 : Learn Data Analysis in Python and R .

Objective

To develop the ability to clean, process, analyze, and visualize data using Python and R, enabling better insights, data-driven decision-making, and efficient handling of real-world datasets.

Tools:

- Python

```
import pandas as pd

df = pd.read_csv("iris.csv")
print(df.head())
print(df.describe())
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
	sepal_length	sepal_width	petal_length	petal_width	
count	150.000000	150.000000	150.000000	150.000000	
mean	5.843333	3.057333	3.758000	1.199333	
std	0.828066	0.435866	1.765298	0.762238	
min	4.300000	2.000000	1.000000	0.100000	
25%	5.100000	2.800000	1.600000	0.300000	
50%	5.800000	3.000000	4.350000	1.300000	
75%	6.400000	3.300000	5.100000	1.800000	
max	7.900000	4.400000	6.900000	2.500000	

I used Python along with the pandas library to perform data analysis on the Iris dataset. I loaded the dataset using read_csv() and stored it in a DataFrame. Then, I displayed the first few records using head() to understand the structure of the data. I also used describe() to generate summary statistics such as mean, standard deviation, minimum, and maximum values. This helped me gain insights into the dataset and understand its key characteristics.

- R

```
data(iris)
head(iris)
summary(iris)
```

Output :

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
Min.	:4.300	Min. :2.000	Min. :1.000	Min. :0.100
1st Qu.:	5.100	1st Qu.:2.800	1st Qu.:1.600	1st Qu.:0.300
Median :	5.800	Median :3.000	Median :4.350	Median :1.300
Mean :	5.843	Mean :3.057	Mean :3.758	Mean :1.199
3rd Qu.:	6.400	3rd Qu.:3.300	3rd Qu.:5.100	3rd Qu.:1.800
Max. :	7.900	Max. :4.400	Max. :6.900	Max. :2.500

	Species
setosa	:50
versicolor	:50
virginica	:50

I used R to perform data analysis on the built-in Iris dataset. I loaded the dataset using `data(iris)` and viewed the first few records using `head()` to understand its structure. I then applied `summary()` to generate statistical information such as mean, median, minimum, and maximum values. This helped me analyze the dataset and understand its key features.

Task 138 : Use API to Fetch Data .

Objective :

To retrieve real-time or remote data from external systems using APIs (such as a REST API), enabling automated data collection, integration with applications, and efficient analysis of up-to-date information for decision-making.

```
import requests

url = "https://jsonplaceholder.typicode.com/posts"
response = requests.get(url)

data = response.json()
print(data[:1])

[{'userId': 1, 'id': 1, 'title': 'sunt aut facere repellat provident occaecati excepturi optio reprehenderit', 'body': 'quia et suscipit  
consequuntur expedita et cum\nreprehenderit molestiae ut ut quas totam\nnostrum rerum est autem sunt rem eveniet architecto'}]
```

I used Python and the requests library to fetch data from a public API. I sent a GET request to the JSONPlaceholder API and received the response in JSON format.

I then converted the response into Python data using `response.json()` and displayed the first record using slicing. This helped me understand how to retrieve and work with real-time data from APIs.

Task 139 : Perform Data Analysis using Jupyter Notebook

Objective :

To analyze, visualize, and interpret data in an interactive environment by writing and executing code step-by-step, enabling better understanding, efficient experimentation, and clear presentation of data insights.

Description :

I used Jupyter Notebook to perform data analysis in an interactive and user-friendly environment. I loaded the dataset and executed Python code step-by-step, which allowed me to understand the data in a structured manner. I used libraries such as pandas to clean and organize the data, including handling missing values and checking data types.

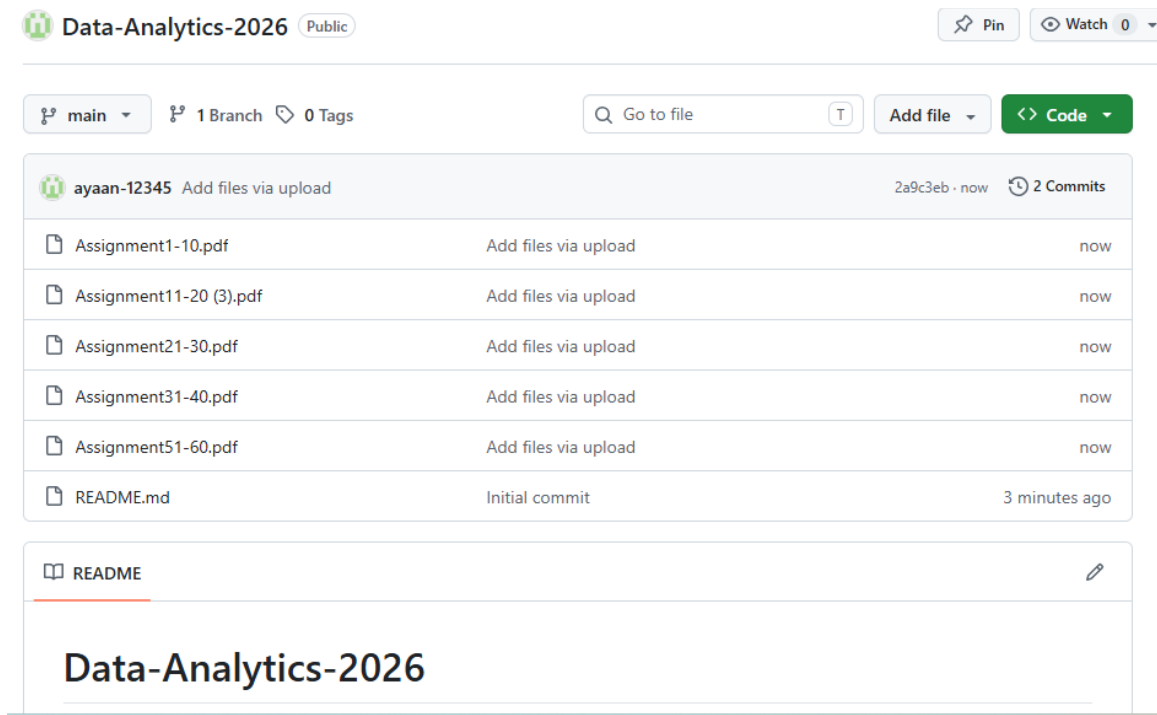
I explored the dataset using functions like `head()`, `info()`, and `describe()` to understand its structure and statistical properties. I also performed basic data manipulation and analysis to identify patterns and trends. Additionally, I used visualization libraries like matplotlib to create simple graphs, which helped in better interpretation of the data.

Jupyter Notebook enabled me to combine code, outputs, and explanations in a single document, making the analysis clear, organized, and easy to present. This improved my understanding of the dataset and helped in drawing meaningful insights.

Task 140 : Collaborate using Git and GitHub .

Objective :

To manage and track changes in code using Git, and collaborate with others through GitHub by sharing repositories, maintaining version history, and enabling efficient teamwork in software and data analysis projects.



I created a GitHub repository named “Data-Analytics-2026” and initialized it with a README file.

I uploaded multiple assignment files (PDFs) to the repository and organized them properly. I performed commits to track changes and maintain version history. I used GitHub as a platform to store and manage my project files in the cloud.

This helped me understand how to use repositories, commits, and file management in GitHub for collaboration and version control.